

MOMIK SHRESTHA

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🔗 PG-Momik

SUMMARY

Software engineer building full-stack products, working increasingly on the AI side of them. Currently on a HIPAA-compliant AI medical scribe platform. Master's thesis on LoRA fine-tuning for text-to-SQL over IATI aid data. Onboarded an engineering team onto Claude Code and agentic development.

TECHNICAL SKILLS

- AI & LLM: LoRA/PEFT fine-tuning, text-to-SQL, local LLM inference, prompt & context engineering, evaluation
- Agentic Development: Claude Code, Devin (Enterprise), Claude Cowork, Skills authoring, harness & workflow configuration
- Backend: Laravel, Express.js, Node.js, Python, Go, REST API, CKAN API
- Frontend: Vue.js, React, React Native, jQuery, SCSS, Tailwind
- Databases: PostgreSQL, MySQL, MongoDB, Redis
- Development Tools: Docker, Git, PHPUnit, Postman, Linux, Typescript, AWS, Cloudflare Workers, D1
- Software Architecture: MVC, Service-Repo Pattern, Factory Pattern, Multi-tenant, Microservices
- Documentation: Markdown, Scrabble, SwaggerUI, Excalidraw, Draw.io

EXPERIENCE

AI Engineer/Senior Software Engineer
Carepilot Inc.

US, Remote
Mar. 2026 – Present

Building full-stack features for a HIPAA-compliant AI medical scribe platform serving US healthcare providers and handling sensitive clinical data.

- Built and maintained EHR integrations (athenahealth/AthenaOne, Everhealth) for bidirectional sync of clinical and patient data, plus backend services across a microservices architecture.
- Work in an AI-native SDLC: Claude Code for implementation, refactors and test generation; Devin (Enterprise) for autonomous ticket-to-PR; Claude Cowork for cross-repo research and planning.
- **Tech stack:** Laravel, Python, Node.js, React, React Native, PostgreSQL, AWS Aurora (MySQL), Docker, Microservices
- **AI-Native Tooling:** Devin (Enterprise), Claude Code, agentic workflow & harness configuration, custom Skills authoring (org and personal)

Software Engineer II (Short term contract)
Sec/Ops Company [NDA]

UAE, Remote
Dec. 2025 – Feb. 2026

Built and maintained features for an enterprise GRC (Governance, Risk & Compliance) platform helping organizations meet a wide range of regulatory compliance frameworks.

- Drove adoption of agentic development: onboarded the engineering team onto Claude Code, authored reusable Skills encoding shared team workflows, and set conventions for agent-assisted delivery.
- Advised the CEO on Claude Code plan and pricing options to support the rollout.
- Delivered secure, scalable product features with cross-functional teams, and provisioned AWS EC2 infrastructure for high availability.
- **Tech stack:** Laravel, PostgreSQL, Redis, Docker, Stored Procedures, AWS
- **AI-Native Tooling:** Claude Code, Skills authoring, team onboarding & enablement

Software Engineer
YoungInnovations Pvt. Ltd.

Nepal, On-Site
Mar. 2023 – Nov. 2025

Took on responsibilities beyond development across client communication, architecture, mentorship, and interim team leadership.

- Acted as interim team lead, running stand-ups, story planning and code reviews, and mentored 3+ junior developers.
- Scoped features, estimated effort, and aligned deliverables with business goals and timelines.
- **Tech stack:** Vue.js, Tailwind, Laravel, PostgreSQL, TypeScript, Node.js, MongoDB, Docker, Postman

Key projects:

IATI Publisher (*Laravel, Vue.js, TypeScript, PostgreSQL, OpenSource*) | github [🔗](#)

- Took ownership after senior team departures; now lead client communication, technical direction, and end-to-end delivery.
- Cut a critical publishing workflow from 20 minutes to 2 using queues, caching, and S3 storage.
- Added OIDC-based SSO (Asgardeo) and integrated the IATI Registry and Validator APIs with detailed error reporting.
- Automated localization via Google Sheets API and Apps Script, and raised PHPUnit coverage from 55% to 71%.

Career Service Center (*Laravel, PostgreSQL, REST APIs*)

- Built multi-tenant REST APIs with strict RBAC, and integrated SpeedExam, Manatal, and Moodle across auth and data-transformation layers.
- Optimized dashboards with PostgreSQL CTEs to sub-300ms response times.

Aidstream (*Laravel, Vue.js, TypeScript, PostgreSQL*)

- Built an XLSX import that halved data import time, and tuned complex SQL with indexes.
- Resolved 30+ client-facing issues and feature requests, backed by PHPUnit coverage on key workflows.

The Only Ever Obsidian Plugin (*TypeScript, Node.js, Serverless, MongoDB, OpenSource*) | github [🔗](#)

- Cross-platform plugin with offline sync on native Obsidian APIs, built with the Obsidian developer community.
- Serverless APIs for the mobile app, plus a companion Chrome extension.

Backend Developer Trainee

YoungInnovations Pvt. Ltd.

*Nepal, On-Site
Sept. 2022 – Mar. 2023*

Contributed to multiple Laravel projects, completing assigned development and testing tickets.

- Built RESTful APIs with optimized performance and security (400ms response time), and automated localization to halve translation effort.
- Closed 30+ tickets with clean, testable code backed by PHPUnit tests.
- **Tech stack:** Laravel, PostgreSQL, PHPUnit, Git, Docker, Postman

Frontend Developer

TofuTechnovations Pvt. Ltd.

*Nepal, On-site
Apr 2020 – Mar. 2021*

Created webpages and maintained legacy web applications.

- Translated 5+ Figma designs into responsive templates and wired them to RESTful APIs.
- Maintained and refactored legacy jQuery codebases.
- **Tech stack:** HTML, CSS, JS, jQuery, SCSS, Tailwind, Figma, Postman

SIDE PROJECTS

BuyMeSutta.com (*Next.js 16, React 19, Tailwind v4, PostgreSQL, eSewa*)

buymesutta.com [🔗](#)

- A micro-saas, creator-support platform, “Buy Me a Coffee” alternative: creators get a page, money moves straight from supporter to creator with nothing held in between.

gmail-local-llm (*Node.js, Local LLM, Gmail API, SQLite*)

<https://github.com/PG-Momik/gmail-local-llm> [🔗](#)

- Runs a local LLM against the Gmail API to classify job-search email (application, interview, rejection) entirely on-device, persisting structured results to SQLite. Local inference, no data leaves the machine.

no-nepali-profanity (*TypeScript, Python, Go, PHP, Dart, OpenSource*)
 mukhxadnahunna.com [🔗](#)

- A profanity filter for English, Romanized Nepali and Devanagari that catches leetspeak, stretched and spelled-out letters and Nepali postpositions, while taking care not to block real names. One lexicon and one matching engine, ported to five ecosystems and published on npm, PyPI, Go modules, Packagist and pub.dev.

IPO Calendar (*Cloudflare Workers, D1, Vue.js, Chrome Extension*)
 ipocalendarnepal.com [🔗](#) | Chrome Web Store [🔗](#)

- Tracks NEPSE IPO openings and closings and puts the offerings a user follows on their Google Calendar. A Chrome extension on an edge API that refreshes NEPSE data every three hours, migrated from a Laravel and EC2 backend to Cloudflare Workers and D1.

dither-art (*TypeScript, SVG, Zero-dependency, npm*)

- Seeded 1-bit artwork drawn from any string: ridges, seas, dunes, planets. The same seed always gives the same picture, so anything can have cover art without storing an image. Zero dependencies, about 6 KB.

Genesis Engine (*Vanilla JS, Design Systems, CSS*)

- A procedurally generated design system generator: mutate a seed, preview the result across real UI, then export the whole system as CSS or Tailwind tokens.

laravel-google-sheet-i18n (*Laravel, PHP, Google Sheets API, Packagist*)

- A Laravel package that pushes a project's language keys up to Google Sheets, drafts them there with =GOOGLETRANSLATE(), and pulls the finished lang files back into the app.

Docu-sphere (*Laravel, Inertia, Vue 3, SQLite, Electron, FrankenPHP*)
<https://github.com/PG-Momik/docu-sphere-beta> [🔗](#)

- An offline-first desktop app built with Laravel, Inertia, Vue 3 and SQLite in Electron, packaged with a FrankenPHP binary. Full-stack web-to-desktop application workflow experiment.

EDUCATION

Lincoln University, Malaysia

2024 - 2026

Master's in Computer Sciences

- **Thesis:** Fine-tuned an LLM with LoRA for natural-language querying of IATI aid data, translating analyst questions into SQL over the IATI schema (text-to-SQL).
- Relevant Coursework: System Design, Data Warehouse and Big Data, Data Analytics, Artificial Intelligence

Tribhuvan University, Nepal

2018 - 2023

Bachelor's in Computer Application

- Relevant Coursework: Web Technologies, Database Management Systems, Software Development, Data Structures and Algorithms
- GPA: 3.36/4.0